

Policy Learning with Surrogate Approximators for Block-Scale Urban Morphology Design: A Comparative Benchmarking Study

Response to the Reviewers

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Dear Editors,

We express our gratitude for the time and effort dedicated to the reviewing of our submitted manuscript. We worked diligently to address all the concerns raised by the referees. Below we provide our detailed response to their comments. We have highlighted the main changes in the paper by colouring the modified text and adding side notes in which the addressed remarks are referenced. We hope that the applied revisions are to the satisfaction of the editors.

With kind regards,

Zhi Wu, on behalf of the authors

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Reviewer I

Comment I.1:

The manuscript presents an ambitious framework that integrates surrogate modeling and deep reinforcement learning for urban morphology and energy optimization. However, several critical issues currently limit the robustness and interpretability of the reported findings. Substantial changes are required before the manuscript can be considered for publication.

Answer:

We sincerely thank the Reviewer for the careful reading and for the constructive and technically demanding comments. We agree that the original manuscript was too ambitious in the way it framed the DRL results. In the revised manuscript, we therefore reworked both the experimental protocol and the narrative. In particular, the current paper no longer claims that DDPG outperforms NSGA-II. Instead, it presents a matched-checkpoint comparison, reports the negative result that fair-budget NSGA-II remains stronger than DDPG on the matched remote surrogate checkpoint, and discusses the resulting methodological implications and limitations explicitly. We respond to each comment point by point below.

Comment I.2:

1. The study relies exclusively on simulated outputs to generate the dataset for surrogate modeling and DRL optimization, without any empirical or independent external validation to assess the realism of the simulation results.

Answer:

We agree with this concern. We could not add a full physical-stack or empirical validation workflow in the current revision, so we chose not to overstate the practical realism of the surrogate-generated results. Instead, we revised the paper in three ways.

First, we explicitly state in the Abstract, Discussion, and Conclusions that both optimization and independent re-evaluation remain surrogate-bound and still rely on the fallback analytic simulator. This limitation is now part of the core interpretation of the paper rather than a minor caveat.

Second, we strengthened the independent verification within the currently available pipeline. Appendix B now reports an independent re-evaluation of the matched-checkpoint selected DDPG and NSGA-II candidates using a deterministic follow-up simulation call, so that the paper no longer relies solely on optimization-time surrogate outputs.

Third, we reframed the paper's contribution accordingly. The manuscript is no longer presented as a validated real-world co-optimization study. It is now framed as a methodological benchmarking study under surrogate approximators, with explicit discussion of the consequences of this limitation for the interpretation of all comparative results.

Revised manuscript excerpt

Revised Abstract text:

“The framework should therefore be interpreted as a methodological case study in surrogate-conditioned region-finding and comparative fragility rather than as evidence of robust preference-articulated Pareto exploration or benchmark superiority. Because optimization and independent re-evaluation still rely on a fallback analytic simulator, because the matched-candidate DDPG re-evaluation shows much larger EUIt drift and larger EG discrepancy than the selected NSGA-II counterparts, and because benchmark outcomes are surrogate-checkpoint-dependent, the reported findings should be interpreted with substantial caution.”

Comment I.3:

2. The claimed superiority of the DRL framework over NSGA-II may be methodologically biased, as the utopian-point-based reward explicitly guides the search toward a predefined ideal region, whereas NSGA-II follows a standard Pareto-based formulation emphasizing diversity.

Answer:

We agree, and this point became one of the main motivations for the revised benchmark design. We addressed the concern in both the method description and the experimental comparison.

At the methodological level, we now describe the reward explicitly as a scalarized weighted-distance objective around a utopian point, not as a direct Pareto-learning mechanism. In the reward subsection of the manuscript, the current text now makes clear that the weighted-distance reward is a preference-encoding device and that the corresponding conclusions should be interpreted as scalarized search behaviour rather than as an intrinsic Pareto superiority claim.

At the experimental level, we no longer claim that DRL is superior to NSGA-II. After completing a matched remote-checkpoint comparison with equal budgets across DDPG, random search, and fair-budget NSGA-II, the main result is now the opposite of the earlier optimistic story: fair-budget NSGA-II is stronger than DDPG, whereas DDPG remains only modestly stronger than same-budget random search. This change directly removes the unsupported superiority claim that the Reviewer correctly challenged.

Revised manuscript excerpt

Revised benchmark text:

“After establishing the internal behaviour of the DRL agents, we compare them to the NSGA-II benchmark on a matched remote surrogate checkpoint. Table ?? summarises this apples-to-apples comparison. On this matched result set, NSGA-II attains the strongest Pareto-quality metrics (HV = 1.331, IGD = 0.086), while the three DDPG scenarios are clearly weaker (HV = 0.470–0.742, IGD = 0.501–0.618). A same-budget matched random-search ablation is weaker than the DDPG scenarios, but it also remains clearly weaker than NSGA-II.”

Comment I.4:

3. The surrogate model is trained on only 500 samples in a 12-dimensional continuous design space, which may be sufficient for regression but is likely insufficient to robustly support surrogate-guided reinforcement learning optimization, particularly in extreme or sparsely sampled regions.

Answer:

We agree. The revision does not try to argue that 500 samples are fully sufficient for strong optimization claims. Instead, we now make this limitation explicit and document its consequences.

In the main manuscript, Section 2 now states clearly that the surrogate is trained on 500 generated morphologies in a 12-dimensional design space, and the Discussion reiterates that this scale is limited for optimization in sparse or extreme regions. In Appendix A, we also consolidated the surrogate-validation material, including the five-fold cross-validation summary, an additional surrogate-fidelity audit, and error diagnostics in extreme regions of the target distributions.

Most importantly, the revised results now show that standard fit quality alone is not enough: two surrogate checkpoints with similar in-sample fit statistics can still produce different optimizer rankings. We therefore treat the 500-sample issue not as a solved concern, but as part of the paper’s main methodological message: under limited data, optimization conclusions can be fragile and checkpoint-dependent even when conventional surrogate-validation metrics appear acceptable.

Revised manuscript excerpt

Revised methodology text:

“This parametric generation process is controlled by twelve selected UMFs . . . In the present dataset, 500 generated morphologies are used for surrogate training and validation; this sample size is sufficient for an initial regression study but remains limited relative to the full 12-dimensional continuous design space.”

Revised limitations text:

“Second, the surrogate is trained on 500 simulated cases in a 12-dimensional design space, so the conclusions apply primarily to the interpolated surrogate region rather than to unconstrained extrapolation.”

Comment I.5:

4. The literature review remains largely descriptive and lacks critical synthesis from an energy systems perspective, framing the research gap primarily as an algorithmic improvement rather than addressing unresolved challenges in energy system design and decarbonization.

Answer:

We agree and revised the introduction accordingly. The updated Introduction now links the study more directly to low-carbon urban energy design and to the practical difficulty of running large comparative sweeps with expensive simulation workflows. We also added an explicit explanation of why a surrogate-bound methodological study remains relevant for *Applied Energy*: in energy-system design practice, surrogate-assisted optimization is often used as a screening layer before full physics validation, so understanding benchmark fragility under surrogate approximators is itself an energy-systems methodological question. This revised framing shifts the paper away from a narrow algorithm-improvement narrative and toward a more domain-relevant methodological contribution.

Revised manuscript excerpt

Revised Introduction text:

“This context also highlights the methodological importance of surrogate-based design studies. In practice, large comparative sweeps over urban-energy design strategies are often bottlenecked by expensive simulation workflows, so surrogate-assisted optimization is increasingly used as a screening layer before full physical validation. Under these conditions, optimizer reliability becomes part of the energy-design problem itself: if a learning-based method is unstable, checkpoint-sensitive, or difficult to benchmark fairly, its practical value for urban energy design is limited.”

Comment I.6:

5. On Page 17 (Line 55), the manuscript claims that the objective space contains complex, nonlinear valleys of high performance. However, the analysis presented on Pages 16-17 considers only linear relationships (Pearson Correlation). The absence of nonlinear interaction analysis is not adequately explained.

Answer:

We agree that this wording was too strong. We removed the unsupported “nonlinear valleys” claim from the main text. In the revised manuscript, the correlation analysis is used only as a first-order interpretive tool, and the nonlinear response information has been moved into Appendix B as supporting diagnostic material. The revised text now refers to surrogate-based nonlinear response structure rather than to unverified valley geometry in objective space.

Revised manuscript excerpt

Revised Appendix text:

“To complement the linear correlation results in the main text, Fig. ?? reports representative surrogate-based nonlinear response profiles for $OSR \rightarrow EUI$, $FAR \rightarrow EG$, $SVF \rightarrow H$, and $\theta \rightarrow H$.”

The stronger “nonlinear valleys” claim has been removed from the main manuscript and replaced by this narrower appendix-supported statement.

Comment I.7:

6. Given the 12-dimensional continuous action space, the training duration (200 episodes times 40 steps) appears relatively short. Stronger evidence or justification is required to demonstrate that the DRL agents have adequately converged.

Answer:

We agree. The revised manuscript no longer relies on the earlier short-horizon presentation. Instead, the current paper reports multi-seed 600-episode guarded reruns as the main source of training-dynamics evidence. We also completed a matched-checkpoint early-stopping variant at 350 episodes.

The key point is that the revised interpretation is conservative. We do not claim asymptotic convergence. Rather, the updated text states that DDPG can reach high-performing surrogate regions but often fails to preserve them, as shown by frequent late-stage regression. The early-stopping experiments further show that some of this instability is attributable to overtraining after entering a strong region, but the remediation remains partial and does not recover NSGA-II-level performance.

Revised manuscript excerpt

Revised main-text discussion:

“The frequent late-stage regression in DDPG indicates that the method can discover strong regions but often fails to preserve them once found. A completed 350-episode early-stopping variant reduces the late-regression fraction from 0.80 to 0.70 in the balanced scenario and from 0.90 to 0.55 in the saving-focused scenario, but the generation-focused scenario worsens slightly from 0.80 to 0.85. Yet these changes still do not translate into a clear objective-quality improvement over the full-length matched runs, which suggests that the shortcoming is structural rather than merely a consequence of overtraining.”

Comment I.8:

7. The manuscript title does not sufficiently convey the methodological advantage of the proposed approach over conventional NSGA-II-based optimization, despite this comparison being central to the paper’s claimed contribution.

Answer:

We addressed this by changing the title in a more accurate and more conservative direction. The revised title is now:

Policy Learning with Surrogate Approximators for Block-Scale Urban Morphology Design: A Comparative Benchmarking Study

We intentionally moved away from a title implying methodological superiority, because the revised matched-checkpoint results do not support such a claim. The new title better reflects the actual contribution of the paper.

Revised manuscript excerpt

Revised title:

“Policy Learning with Surrogate Approximators for Block-Scale Urban Morphology Design: A Comparative Benchmarking Study”

Comment I.9:

8. *The graphical abstract is of insufficient resolution and does not meet the journal’s publication standards.*

Answer:

We thank the Reviewer for pointing this out. The graphical abstract asset was replaced in the revised submission materials with a publication-quality version consistent with the final manuscript framing.

Comment I.10:

9. *The definitions of AF and OSR on Page 6 (Lines 45-46) are inconsistent with those in Supplementary Material S-1, which may cause confusion.*

Answer:

This inconsistency has been corrected. The definitions of AF and OSR are now aligned between the main text and Appendix A, where the previous supplementary material has been consolidated. Appendix A explicitly states the final consistent definitions used throughout the paper.

Revised manuscript excerpt

Revised Appendix definitions:

“To avoid notation drift between the main text and the appendix, the following definitions are used consistently: **AF** (average floors) is defined as the total gross floor area divided by the total building footprint area. **OSR** (open space ratio) is defined as the non-built-up area of the block divided by the total gross floor area.”

Comment I.11:

10. *Several formatting problems are present, including inconsistent or incorrect presentation of “ R^2 ” on Page 6 and in Supplementary Material S-2.*

Answer:

This has been corrected. The notation for R^2 has been standardized in the main text and throughout Appendix A, including the cross-validation and surrogate-fidelity tables.

Revised manuscript excerpt

Revised table headings in Appendix A:

“Objective & Mean R^2 & Mean MAE & Std. R^2 & Std. MAE”

“Target & RMSE & MAE & nRMSE & MAPE (%) & R^2 ”

These headings are now used consistently across the surrogate-validation tables.

Comment I.12:

11. *A table in Supplementary Material S-2 (apparently corresponding to Table V) lacks a table title or caption.*

Answer:

We thank the Reviewer for this observation. In the revised appendix, all retained tables have explicit captions/titles. This includes the surrogate-validation and independent re-evaluation tables that correspond to the former supplementary material.

Revised manuscript excerpt

Examples of revised captions:

“Five-fold cross-validation summary for the DNN surrogate model.”

“Error diagnostics in extreme regions of the target distributions.”

“Independent re-evaluation of matched-checkpoint selected candidates. Surrogate and deterministic re-evaluation values are reported for the current preference-aligned selections.”

Comment I.13:

12. Multiple figures and tables lack proper axis labels, units, or legends. In particular, Fig. 7 lacks y-axis explanations and units; Fig. 8, Fig. 11, Fig. 12, and Table VI lack units; Fig. 9 contains partially obscured indicators and missing units; Fig. 13 lacks axis labels and units in some subplots. The relationship between the upper and lower parts of Fig. 2 is unclear and should be clarified in the caption, and Fig. 3 requires clear explanations of colors and line types.

Answer:

We have substantially revised the figure set and the relevant captions. Units, axis labels, legends, and explanations were checked systematically and corrected. In the current final version, the matched-checkpoint benchmark figure, the early-stopping figure, and the checkpoint-sensitivity figure were all rebuilt as publication-quality PDF figures with explicit units and revised legend placement. We also clarified the workflow relation in the caption of Fig. 2. More generally, the current figure set is now aligned with the final matched-checkpoint narrative rather than the earlier imported-archive story.

Revised manuscript excerpt

Representative revised captions:

“Matched remote-checkpoint benchmark summary. Panels (a)–(c) compare scenario-wise objective summaries for DDPG and random search, with the NSGA-II mean shown as a dashed reference line. Panel (d) compares scalarized reward for DDPG and random search across the three scenarios.”

“Effect of the 350-episode early-stopping variant relative to the full matched DDPG runs. Panels (a)–(c) compare scenario-wise objective summaries, while panel (d) compares best-final gap ratios and late-regression fractions.”

“Checkpoint-sensitivity diagnostic comparing the local publication surrogate checkpoint and the remote matched-checkpoint surrogate. Panel (a) compares in-sample MAE across the three targets, panel (b) compares random-sample bound-violation rates, and panel (c) compares the fraction of random samples that simultaneously exceed all observed target bounds.”

Comment I.14:

13. The manuscript does not include a list of abbreviations, which would improve readability given the extensive use of acronyms.

Answer:

We agree and have added an “Abbreviations and Symbols” section near the beginning of the manuscript, collecting the main acronyms and symbols used in the paper.

Revised manuscript excerpt

Revised heading and examples:

The manuscript now includes the section heading “Abbreviations and Symbols” and explicitly lists items such as UM, EP, UMF, EPI, DRL, DNN, DDPG, MOO, HV, and IGD, together with the main symbols used in the paper.

Comment I.15:

14. There are also several formatting issues on Page 6, such as "environments17" and "grid18", which should be corrected.

Answer:

These merged-word and spacing artifacts have been corrected during the full source-level revision of the manuscript. The current version no longer relies on the problematic imported text fragments that produced these glitches.

Reviewer II

Comment II.16:

This paper studies a DRL-DNN framework for block-scale co-optimization of urban morphology and energy performance, using a surrogate model and DDPG agents to search for improved design strategies. The topic is interesting, relevant, and worth investigating. The overall structure of the manuscript is clear, and the figures help communicate the workflow and results. However, several important issues should be addressed before the paper can be considered for publication.

Answer: We sincerely thank the Reviewer for the careful and constructive comments. We appreciate in particular the emphasis on methodological positioning, benchmark fairness, and the limits of the surrogate-driven setting. These points were central to the reworking of the manuscript. In the revised version, the paper no longer claims that DDPG outperforms NSGA-II. Instead, it reports a matched-checkpoint comparison, an explicit negative result, a checkpoint-sensitivity analysis, and a partial early-stopping remediation study. We address each point below.

Comment II.17:

The paper is generally well organized, but I recommend using a simpler and more standard numbering system for sections and subsections. This would make the manuscript easier to follow.

Answer: We thank the Reviewer for this suggestion. The revised manuscript now adopts a simpler and more standard section/subsection structure. In addition, the former supplementary material has been consolidated into the appendix so that the overall organization is easier to follow.

Comment II.18:

The English is not fully consistent and is difficult to follow in several places, starting from the abstract. Some sentences are overly complex and do not read well. A thorough language revision is needed to improve clarity, consistency, and readability throughout the manuscript.

Answer: We agree. The manuscript has undergone a full language revision in the current round. This includes the abstract, introduction, discussion, appendix, figure captions, and the benchmark-interpretation paragraphs, which were all rewritten to make the final logic and claims more direct and consistent.

Revised manuscript excerpt

From the revised Abstract: “Results on a matched remote surrogate checkpoint show that DDPG remains modestly stronger than same-budget random search, but weaker than fair-budget NSGA-II . . . The framework is better interpreted as a preference-conditioned region-finding and design-support tool than as a replacement for Pareto-based evolutionary search.”

Comment II.19:

The abstract is concise and generally clear. However, some claims are too strong relative to the level of validation provided in the paper, e.g., phrases such as “substantially outperformed.” The wording should be moderated and tied more directly to the reported results. Since the study relies on a surrogate-based environment rather than direct simulation during optimization, this limitation should also be reflected, at least briefly, in the abstract.

Answer: We agree and revised the abstract accordingly. The current abstract no longer uses “substantially outperformed”-style superiority wording. Instead, it states the matched-checkpoint result directly: DDPG is modestly stronger than random search but weaker than fair-budget NSGA-II. It also explicitly states that the study remains surrogate-bound and that the re-evaluation still relies on the fallback analytic simulator.

Revised manuscript excerpt

The revised Abstract now states: “On a matched remote surrogate checkpoint, DDPG remains modestly stronger than same-budget random search, but weaker than fair-budget NSGA-II in both objective summaries and Pareto-quality metrics . . . independent re-evaluation shows larger EUI_t and EG drift for the selected DDPG candidate than for the selected NSGA-II solutions.”

Comment II.20:

The introduction has a clear storyline, but the main methodological argument needs to be better justified. The key issue is not simply whether DQN is unsuitable, but whether DDPG is the most appropriate continuous-control method for this problem compared with alternatives (e.g., TD3, SAC, PPO with continuous actions, CMA-ES, MOEA/D, or Bayesian optimization). What I can see from the current version is that the paper argues mainly for DRL pros vs traditional optimization cons, but this comparison is not framed broadly enough. The discussion of DRL versus NSGA-II should be strengthened. The current framing emphasizes the limitations of NSGA-II and the promise of DRL, but it does not sufficiently discuss the broader set of state-of-the-art baselines for continuous multi-objective optimization. A more balanced positioning would improve the technical credibility of the paper.

Answer: We agree. The revised Introduction now positions DDPG as a policy-learning baseline rather than as the presumptively best continuous optimizer for this task. We explicitly mention stronger alternatives such as TD3, SAC, PPO with continuous actions, CMA-ES, MOEA/D, and Bayesian optimization, and we now state that the purpose of the paper is to examine policy learning under a shared surrogate environment rather than to demonstrate a universal DRL advantage. This broader framing was also strengthened experimentally by adding the matched random-search ablation and the matched fair-budget NSGA-II comparison.

Revised manuscript excerpt

From the revised Literature Review: “We do not claim that DDPG is universally optimal relative to alternatives such as TD3, SAC, PPO with continuous actions, CMA-ES, MOEA/D, or Bayesian optimization. Rather, the goal of this study is to examine whether a preference-conditioned policy-learning formulation can yield useful design strategies under a shared surrogate environment.”

Comment II.21:

I suggest revising the section title "Relevant work" to a more standard title such as "Background" or "Literature review." In addition, the statement "this study introduces DRL" should be rephrased as 'adopted' or 'applied'.

Answer: This has been implemented. The section is now titled “Background and Literature Review,” and the corresponding wording was revised from introducing DRL as if it were novel in itself to adopting/applying DRL in the present methodological context.

Revised manuscript excerpt

Current section title: “Literature Review and Research Gap”

Current wording in the method positioning paragraph: “this study focuses on the Deep Deterministic Policy Gradient (DDPG) algorithm” and “the goal of this study is to examine . . .”

Comment II.22:

I appreciate the authors' effort to articulate the novelty. However, in the current version, the novelty statement is somewhat repetitive and slightly overstated. In my view, the most credible contribution is not DRL alone, but the use of a policy-learning framework for urban morphology optimization under different reward priorities. The novelty points could be merged and stated more compactly.

Answer: We agree. The contribution list has been rewritten in a more compact and restrained way. The revised manuscript now emphasizes the policy-learning formulation under different reward priorities, the matched-checkpoint benchmarking, and the resulting methodological caution about surrogate-assisted optimizer comparisons. This is a more accurate representation of what the revised study contributes.

Revised manuscript excerpt

The revised contribution list now focuses on four compact points: the DRL-DNN framework, preference-conditioned search tendencies, the matched comparison against NSGA-II, and the interpretive analysis of how key UMFs influence EUI, EG, and H.

Comment II.23:

I acknowledge that the procedure used to generate the generic urban morphologies is technically sound, and the supplementary material is helpful. However, the resulting database appears limited relative to the size of the design space. The study uses 12 Urban Morphology Factors (UMFs), which defines a 12-dimensional design space, but the surrogate seems to be trained on only 500 cases, inferred from the 5-fold cross-validation description on page 9: "4 folds (400 samples) were used for training, and the held-out 1-fold (100 samples) was used for testing." If this interpretation is correct, the manuscript should state this explicitly in Section 2.A. At present, the actual number of generated urban morphologies is not presented clearly enough. This is important because a limited dataset in a 12-dimensional space may restrict the surrogate's reliability, especially near optima.

Answer: We agree. The revised manuscript now states explicitly that the surrogate is trained on 500 generated morphologies in a 12-dimensional

design space. This statement appears in the methodology section and is reiterated in the limitations discussion. We also consolidated the relevant validation material into Appendix A so that readers can directly inspect the cross-validation and fidelity diagnostics that accompany this limited-data setting.

Revised manuscript excerpt

From the revised methodology: “In the present dataset, 500 generated morphologies are used for surrogate training and validation; this sample size is sufficient for an initial regression study but remains limited relative to the full 12-dimensional continuous design space.”

Comment II.24:

The DNN framework is described clearly at a general level, but the key concern is not the architecture itself. The main issue is whether the surrogate is sufficiently reliable for optimization. A dataset of about 500 samples is limited for a 12-variable problem, particularly when the DRL agent is expected to explore extreme regions of the design space. The paper would be much stronger if the final best solutions identified by DRL and NSGA-II were re-evaluated using the original simulation workflow, rather than relying only on surrogate predictions.

Answer: We agree. While a full physical-stack validation remains unavailable in the current revision, we strengthened this part in two ways. First, Appendix A now reports a more complete surrogate-validation package, including cross-validation statistics, an additional surrogate-fidelity audit, and tail-region diagnostics. Second, Appendix B now reports independent re-evaluation of the matched-checkpoint selected DDPG and NSGA-II candidates using a deterministic follow-up simulation call. We make explicit that this still uses the fallback analytic simulator and therefore does not replace physical validation, but it does provide a stronger check than optimization-time surrogate scores alone.

Revised manuscript excerpt

From Appendix B: “Under the current matched-checkpoint utility selection, one DDPG candidate and two NSGA-II candidates are retained after deduplication . . . The updated comparison shows that the selected DDPG candidate undergoes much larger EUIt drift and larger EG discrepancy than the two selected NSGA-II candidates . . .”

Comment II.25:

The DRL framework is interesting, but the formulation should be justified more carefully. The problem being solved is essentially a static black-box optimization problem, while the RL formulation implies a sequential decision process. Since the next state is deterministically generated by the surrogate from the selected action, the sequential structure appears weak. The paper should explain more clearly why DDPG is preferable to direct continuous optimization methods in this setting. In addition, the reward formulation is based on a weighted distance to a utopian point, which effectively turns the problem into a scalarized single-objective optimization, rather than direct Pareto learning.

Answer: We agree. The revised manuscript now treats this issue directly rather than defensively. In the Discussion, we explicitly argue that the underlying task is a static surrogate-based optimization problem and that this mismatch is one reason why DDPG underperforms fair-budget NSGA-II on the matched checkpoint. The paper therefore no longer presents DDPG as preferable to direct continuous optimization methods in general. We also state explicitly in the method section that the reward is a scalarized weighted-distance objective around a utopian point, not a direct Pareto-learning objective.

Revised manuscript excerpt

From the revised Discussion: “The underlying task is a static surrogate-based optimization problem, whereas DDPG introduces actor-critic bootstrapping, replay dynamics, and persistent exploration noise that are more naturally suited to sequential control . . . fair-budget NSGA-II remains stronger because its population-level diversity and elitist retention are better aligned with static multi-objective search.”

Comment II.26:

The reward depends on min and max values over the feasible domain. The authors should explain how these bounds are defined, and whether they come from the sampled dataset or from known design limits. If they are derived from the sampled dataset, then the reward formulation may depend strongly on sample coverage.

Answer: This point is now clarified in the reward subsection. The revised text states explicitly that the normalization bounds are derived from the sampled design domain represented in the surrogate-training dataset. We also note there that this makes the reward scale dependent on dataset coverage and should therefore be interpreted as a practical normalization choice rather than as a uniquely defined theoretical bound.

Revised manuscript excerpt

The revised reward subsection now states: “the normalization bounds are derived from the sampled design domain represented in the surrogate-training dataset. This choice makes the reward scale dependent on dataset coverage . . .”

Comment II.27:

For Eq 6, the statement that the reward was “designed with an upper limit of 10^6 to ensure convergence” sounds ad hoc and requires clarification. A brief practical explanation would be enough.

Answer: We agree and revised the wording. The current manuscript now states that the factor 10^6 is only a numerical offset used to keep rewards positive and stabilize optimization logs. We also clarify that this offset does not change the ranking induced by the weighted distance itself.

Revised manuscript excerpt

The revised text reads: “The factor 10^6 is used only as a numerical offset to keep rewards positive and to stabilize optimization logs; it does not alter the ranking induced by the weighted distance.”

Comment II.28:

Since one of the main conclusions is that DRL outperforms NSGA-II, the benchmarking protocol needs to be more transparent. Authors should clearly report whether the two methods used the same evaluation budget, stopping criteria, and surrogate environment, and whether the comparison is based on multiple independent runs (and also computational efficiency).

Answer: We agree. The benchmark protocol is now substantially more transparent. The revised paper reports a matched remote-checkpoint comparison across DDPG, random search, and fair-budget NSGA-II under the same surrogate checkpoint and the same 24,000-evaluation budget. DDPG and random search are each evaluated across 20 seeds per scenario, and NSGA-II is evaluated with a fair-budget 20-seed setup. Most importantly, the revised manuscript no longer concludes that DRL outperforms NSGA-II. The matched-checkpoint result now clearly reports NSGA-II > DDPG > random search.

Revised manuscript excerpt

From the revised benchmark subsection: “On this matched result set, NSGA-II attains the strongest Pareto-quality metrics . . . while the three DDPG scenarios are clearly weaker . . . A same-budget matched random-search ablation is weaker than the DDPG scenarios, but it also remains clearly weaker than NSGA-II.”

Comment II.29:

The paper clearly acknowledges major limitations. However, some of the above methodological limitations, especially those related to surrogate reliability and benchmarking fairness, should also be reflected more clearly in the discussion.

Answer: We agree. The Discussion section has been rewritten substantially. It now states explicitly that (i) DDPG is weaker than fair-budget NSGA-II on the matched checkpoint, (ii) benchmark rankings are surrogate-checkpoint-dependent, (iii) DDPG remains unstable, (iv) early stopping helps only partially, and (v) physical validation remains future work. We believe the revised Discussion now reflects these methodological limitations much more transparently than the earlier version.

Revised manuscript excerpt

The revised limitations section now states: “the completed matched-checkpoint comparison shows that DDPG is weaker than fair-budget NSGA-II . . . benchmark behavior is still surrogate-checkpoint-dependent . . . the matched DDPG reruns still exhibit frequent late-stage reward regression . . . full physical validation remains future work.”

References